

Paper And Boards

Choosing the right board

Each board is special chosen and designed for its intended purpose.
Boards are usually used for:

- Packaging
- Drawing and sketching
- Model making





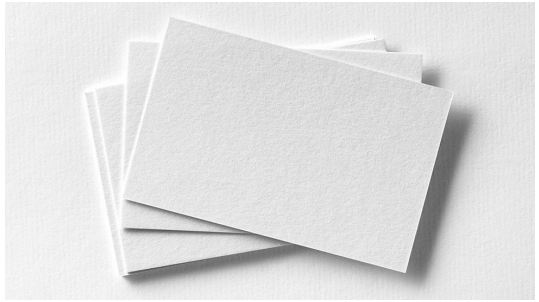




However companies also need the material to fit a certain criteria on its ability to be recycled, its cost and specific properties needed for certain tasks. (Thickness and rigidity for printing)

Papers and boards can have a range of properties:

- Textured
- Embossed
- Printed on
- Laminated (Making it waterproof)



https://www.bbc.co.uk/bitesize/guides/znq8jty/video

Corrugated board



Corrugated board is a robust and protective material made from two outer layers of flat paperboard (Liners) with a rippled layer of paper sandwiched between them (Flute)

This layered structure, inspired by the strength of arches, gives the board its rigidity, cushioning properties. Making it ideal for packaging and shipping.

Strengths

Printable: The surface of corrugated board can be easily printed on for branding and marketing purposes.

Strength and protection: The arch-like structure of the flutes provides high resistance to impact, vibration, and crushing.

Versatile: Can be cut and folded into different sizes and shapes for various packaging applications.

Lightweight: Its construction makes it light and easy to handle, which helps minimize shipping costs.

Insulation: The air trapped in the flutes acts as an insulator, protecting contents from temperature fluctuations.

Cost-effective: It is an affordable and practical material for a wide variety of packaging needs.

Eco-friendly: Corrugated board is biodegradable and made from renewable resources. It is also highly recyclable, with over 80% of corrugated packaging in the UK being recycled.

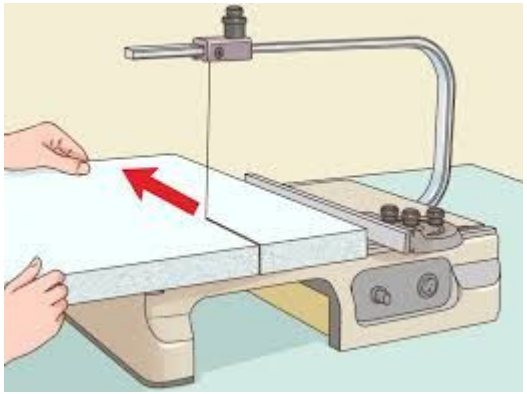
Foam Board



Foam board is a lightweight and versatile material used for a wide range of applications, from arts and crafts to professional presentations. It consists of a dense foam core, typically made from polystyrene, with an outer facing of bleached paper, clay coated paper, or vinyl.

The most common uses for this board is for:

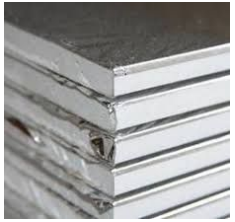
- Mounting artwork, providing a stiff backing for poster prints and photographs
- Architectural models, rigid but easily cut for structure
- Presentation boards,
- Signage
- Craft projects



There are a range of different ways to cut this material such as:

- Craft knives
- Hot wires or knives, using electricity to heat up the blade as well as leaving minimal debris behind after cutting
- Saw, creates a large amount of foam sudt which sticks to many things due to static created

Foil lined board



Foil lined board is mostly used where there is a need for a moisture barrier, thermal insulation or for better aesthetics. It is great for certain packaging as well as some construction.

In food packaging it is perfect due to its ability to act as a barrier against oxygen, light helping to reserve the food for a longer period of time.
Examples of it being used in food places can be

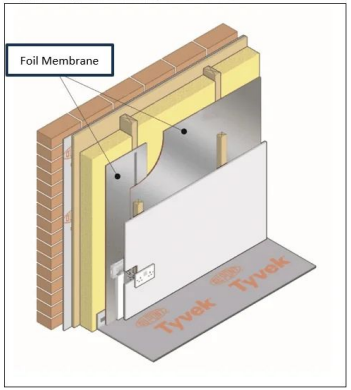
- Take away containers, maintaining the temperature during transport
- Ready meals
- Drinks cartons, allowing milk to sit at room temperature for longer than usual whilst still staying safe

In construction it is great for insulation and moisture control

Being a vapour barrier it prevents moisture from passing through walls and ceilings in your home. It also helps reduce condensation levels which can cause water damage over a long period of time or even mould growth

The foil lining allows for heat to be trapped or repelled keeping you home cool in summer and warm in winter. It can be seen as an energy efficient product too.

Figure 1 Typical wall and floor installation











Folding box board



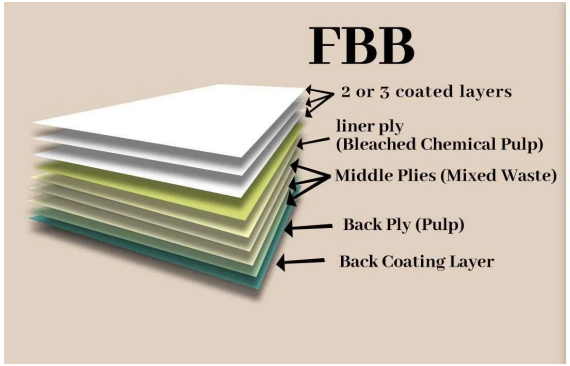
Folding boxboard (FBB)is a paper board used for high quality packaging and printing. It consists of a pulp sandwiched between two outer layers of bleached pulp. This creates a low density material with great stiffness as well as a smooth finish.

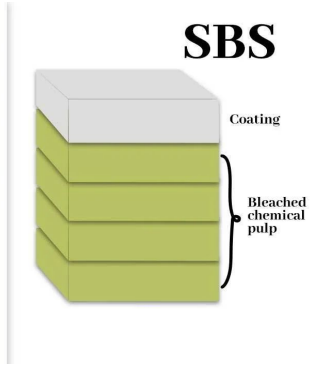


Folding box board has a range of advantages such as its:

- Versatility as it can be used and changed to fit many different purposes with uses of finishes and coatings
- Light weight material makes it perfect for shipping, with less weight costs
- Cost effectiveness of the materials used to make it
- Great strength

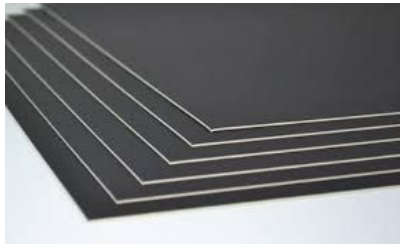
This material is also very foodsafe making it perfect for food packaging





Mounting boards

Also known as backing boards or foam core, it provides a ridgid, sturdy foundation for framed art work, photos and even presentations.



Depending On the material used it can even provide some protection against acid damages, but mainly stops the material from moving or warping over time.

It is typically made from polystyrene foam centre sandwiched between paper creating an economical lightweight material

- **Lightweight** Despite its rigidity, foam board is very light, making it easy to transport, handle, and install, even in large sizes. This is a major benefit for trade or presentations where mobility would be ideal.
- **Cost Effective** Foam boards are a budget-friendly alternative to more durable materials
- **Insulating Properties** The foam core provides good thermal and sound insulation, which can be a valuable benefit in architectural and construction

